HW9

Problem 1

```r
#load libraries
library(nnet)
library(lmtest)
```

```
## Warning: package 'lmtest' was built under R version 4.3.3

## Loading required package: zoo

## Warning: package 'zoo' was built under R version 4.3.3

##
## Attaching package: 'zoo'

## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric
```

```r
#create a list
disease <- c(1167, 1509, 763, 234, 107, 14, 377, 16, 225, 2, 29, 0, 186, 179, 130, 19, 11, 2)
#create an array
dis.array <- array(disease, c(2,3,3))
#saturated log-linear model
sat.log = loglin(dis.array, margin = list(c(1,2)), fit = TRUE, param = TRUE)
```

```
## 2 iterations: deviation 0
```

```r
#check p-value for testing the null hypothesis for G^2
1-pchisq(sat.log$lrt, sat.log$df)
```

```
## [1] 0
```

```r
sat.log
```

```
## $lrt
## [1] 4446.676
##
## $pearson
## [1] 4446.556
```

```
##
## $df
## [1] 12
##
## $margin
## $margin[[1]]
## [1] 1 2
##
##
## $fit
## , , 1
##
##          [,1]     [,2]      [,3]
## [1,] 576.6667 372.6667 49.000000
## [2,] 568.0000  85.0000  5.333333
##
## , , 2
##
##          [,1]     [,2]      [,3]
## [1,] 576.6667 372.6667 49.000000
## [2,] 568.0000  85.0000  5.333333
##
## , , 3
##
##          [,1]     [,2]      [,3]
## [1,] 576.6667 372.6667 49.000000
## [2,] 568.0000  85.0000  5.333333
##
##
## $param
## $param$`(Intercept)`
## [1] 4.77142
##
## $param$`1`
## [1]  0.6185033 -0.6185033
##
## $param$`2`
## [1]  1.5782732  0.4102481 -1.9885213
##
## $param$`1.2`
##            [,1]       [,2]       [,3]
## [1,] -0.6109318  0.1205132  0.4904186
## [2,]  0.6109318 -0.1205132 -0.4904186
```

We return a 1, which tells us that the saturated model does fit the data well. We will look at other models as well, but based on the saturated log-linear model we know that we have statistically significant evidence to suggest that there is a relationship between SNP1 and SNP 2 based on if they have the disease. We can also check for the independence of specific relationships (such as X1 to X2 and 3):

```
x1indep = loglin(dis.array, margin = list(1, c(2,3)), fit = TRUE, param = TRUE)
```

```
## 2 iterations: deviation 4.547474e-13
```

```
#check p-value for testing the null hypothesis for G^2
1-pchisq(x1indep$lrt, x1indep$df)
```

```
## [1] 0
```

```
x1indep
```

```
## $lrt
## [1] 1052.485
##
## $pearson
## [1] 892.8751
##
## $df
## [1] 8
##
## $margin
## $margin[[1]]
## [1] 1
##
## $margin[[2]]
## [1] 2 3
##
##
## $fit
## , , 1
##
##        [,1]     [,2]    [,3]
## [1,] 1612.6 600.8078 72.9165
## [2,] 1063.4 396.1922 48.0835
##
## , , 2
##
##         [,1]      [,2]     [,3]
## [1,] 236.828 136.79376 17.47586
## [2,] 156.172  90.20624 11.52414
##
## , , 3
##
##          [,1]     [,2]     [,3]
## [1,] 219.9547 89.78974 7.834004
## [2,] 145.0453 59.21026 5.165996
##
##
## $param
## $param$`(Intercept)`
## [1] 4.5995
##
## $param$`1`
## [1]  0.2081879 -0.2081879
##
## $param$`2`
## [1]  1.2744320  0.4637192 -1.7381512
```

```
##
## $param$`3`
## [1]  1.2167101 -0.3921446 -0.8245655
##
## $param$`2.3`
##               [,1]       [,2]       [,3]
## [1,]  0.086773182 -0.2226409  0.1358677
## [2,] -0.089841705  0.0392123  0.0506294
## [3,]  0.003068524  0.1834286 -0.1864971
```

Based on the G^2 equal to zero, we can see that this does not fit the data correctly, indicating that X1 is not independent. What this means for our data is that if someones has the disease, this does tell us some information about X2 and X3.

We can then do the same thing for both X2 and X3:

```r
#for x2 being independent
x2indep = loglin(dis.array, margin = list(2, c(1,3)), fit = TRUE, param = TRUE)
```

```
## 2 iterations: deviation 4.547474e-13
```

```r
#check p-value for testing the null hypothesis for G^2
1-pchisq(x2indep$lrt, x2indep$df)
```

```
## [1] 0
```

```r
x2indep
```

```
## $lrt
## [1] 502.9381
##
## $pearson
## [1] 464.365
##
## $df
## [1] 10
##
## $margin
## $margin[[1]]
## [1] 2
##
## $margin[[2]]
## [1] 1 3
##
##
## $fit
## , , 1
##
##          [,1]     [,2]     [,3]
## [1,] 1407.456 562.7366 66.80704
## [2,] 1213.992 485.3845 57.62394
##
```

```
## , , 2
##
##           [,1]       [,2]       [,3]
## [1,] 435.98672 174.318511 20.6947686
## [2,]  12.43702   4.972636  0.5903421
##
## , , 3
##
##          [,1]     [,2]      [,3]
## [1,] 225.9392 90.33622 10.724547
## [2,] 138.1891 55.25151  6.559356
##
##
## $param
## $param$`(Intercept)`
## [1] 4.228245
##
## $param$`1`
## [1]  0.6994079 -0.6994079
##
## $param$`2`
## [1]  1.3214862  0.4047585 -1.7262447
##
## $param$`3`
## [1]  1.6258730 -1.2505865 -0.3752865
##
## $param$`1.3`
##            [,1]      [,2]       [,3]
## [1,] -0.6254727  1.079059 -0.4535865
## [2,]  0.6254727 -1.079059  0.4535865
```

```r
#for x3 being independent
x3indep = loglin(dis.array, margin = list(3, c(1,2)), fit = TRUE, param = TRUE)
```

```
## 2 iterations: deviation 4.547474e-13
```

```r
#check p-value for testing the null hypothesis for G^2
1-pchisq(x3indep$lrt, x3indep$df)
```

```
## [1] 0
```

```r
x3indep
```

```
## $lrt
## [1] 582.7652
##
## $pearson
## [1] 446.1811
##
## $df
## [1] 10
##
```

```
## $margin
## $margin[[1]]
## [1] 3
##
## $margin[[2]]
## [1] 1 2
##
##
## $fit
## , , 1
##
##          [,1]     [,2]      [,3]
## [1,] 1320.648 853.4592 112.21690
## [2,] 1300.800 194.6620  12.21408
##
## , , 2
##
##          [,1]      [,2]      [,3]
## [1,] 225.9095 145.99235 19.195775
## [2,] 222.5143  33.29879  2.089336
##
## , , 3
##
##          [,1]      [,2]      [,3]
## [1,] 183.4427 118.54849 15.587324
## [2,] 180.6857  27.03924  1.696579
##
##
## $param
## $param$`(Intercept)`
## [1] 4.35346
##
## $param$`1`
## [1]  0.6185033 -0.6185033
##
## $param$`2`
## [1]  1.5782732  0.4102481 -1.9885213
##
## $param$`3`
## [1]  1.2465730 -0.5191704 -0.7274026
##
## $param$`1.2`
##            [,1]       [,2]       [,3]
## [1,] -0.6109318  0.1205132  0.4904186
## [2,]  0.6109318 -0.1205132 -0.4904186
```

Both of these models return us a p-value of 0 based on our G^2 statistic. What this means is that neither X2 or X3 are independent. This means that knowledge about SNP1 gives us information about if the person has the disease and what type of SNP 2 the person has. This also means that the knowledge about what type of SNP 2 the person has it tells us something about if they have the disease and what type of SNP 1 they have.

We can now move on to models of conditional independence:

```r
x2indepX3givenX1 = loglin(dis.array, margin = list(c(1,2), c(1,3)), fit = TRUE, param = TRUE)
```

```
## 2 iterations: deviation 2.273737e-13
```

```r
1-pchisq(x2indepX3givenX1$lrt, x2indepX3givenX1$df)
```

```
## [1] 0.552064
```

```r
x2indepX3givenX1
```

```
## $lrt
## [1] 6.857671
##
## $pearson
## [1] 6.354666
##
## $df
## [1] 8
##
## $margin
## $margin[[1]]
## [1] 1 2
##
## $margin[[2]]
## [1] 1 3
##
##
## $fit
## , , 1
##
##          [,1]     [,2]     [,3]
## [1,] 1176.631 760.3893 99.97963
## [2,] 1515.913 226.8532 14.23392
##
## , , 2
##
##           [,1]       [,2]       [,3]
## [1,] 364.48414 235.545242 30.9706177
## [2,]  15.53013   2.324051  0.1458228
##
## , , 3
##
##          [,1]      [,2]      [,3]
## [1,] 188.8848 122.06544 16.049750
## [2,] 172.5570  25.82278  1.620253
##
##
## $param
## $param$`(Intercept)`
## [1] 3.992946
##
## $param$`1`
```

```
## [1]  1.109723 -1.109723
##
## $param$`2`
## [1]  1.5782732  0.4102481 -1.9885213
##
## $param$`3`
## [1]  1.6258730 -1.2505865 -0.3752865
##
## $param$`1.2`
##            [,1]       [,2]       [,3]
## [1,] -0.6109318  0.1205132  0.4904186
## [2,]  0.6109318 -0.1205132 -0.4904186
##
## $param$`1.3`
##            [,1]      [,2]       [,3]
## [1,] -0.6254727  1.079059 -0.4535865
## [2,]  0.6254727 -1.079059  0.4535865
```

We can see that this is quite high, which means that this does fit the data well. This makes complete sense since if a simple model fits the dataset a more complex model that embeds it is also likely to fit the dataset. And we know that our saturated log-linear model is nested in the conditional independence model above.